

Final Project Report

MACS 40550: Agent-Based Modeling

Spatial Structure and the Mobilization Consequences of Repression: An Agent-Based Model of Protest Dispersal

An Nisa Astuti (annisaastuti@uchicago.edu)
<https://github.com/asstanns/abm-spatial-repression>

1. Introduction

When state repression disperses a protest crowd, what determines whether dispersed individuals reconstitute into a mobilized collective or demobilize permanently? This paper argues that the built environment surrounding a protest site shapes the dispersal pattern of protest after repression. Existing scholarship has disaggregated repression by type, showing that lethal violence tends to backfire while routine intervention tends to deter (Davenport 2007; Sutton, Butcher, and Svensson 2014). However, the repression-mobilization literature documents diverging outcomes even after this disaggregation, showing that there are diverging outcomes in which the sources of variation are not explicitly specified (Davenport 2007; Earl 2011). I argue that spatial structure of the urban environment contribute to the diverging mechanism in this within-type repression response variation.

The contribution of this paper sits at the intersection of three bodies of literature that still largely disconnected. The repression-mobilization literature has progressed in identifying when repression backfires versus deters but treats space only as a unit of statistical aggregation. The threshold model tradition in collective action has developed sophisticated micro-foundations for cascade dynamics but assumes aspatial observation. Moreover, the spatial turn in contentious politics literature has established that physical space matters substantively but has rarely formalized the mechanism. This model contributes to these debates by spatializing the threshold model in a way that connects observable urban morphology to mobilization outcomes under repression.

2. Literature Review

2.1 Repression and its mobilizational consequences

The relationship between state repression and mobilization is one of the most contested questions in contentious politics. Davenport (2007) surveys 40 years of repression research and identifies a central puzzle: while dissent consistently provokes repression through 'law of coercive responsiveness', the reverse relationship is deeply inconsistent. Repression sometimes deters dissent, sometimes escalates it, and sometimes has no effect at all. This inconsistency, which Davenport terms the 'punishment puzzle,' has driven efforts to identify the conditions under which repression produces each outcome. Lichbach (1987) identified this

inconsistency early, framing it as the deterrence-or-escalation puzzle in aggregate studies of repression and dissent. Sutton, Butcher, and Svensson (2014) show empirically that even among cases of extreme repression against unarmed protesters, backfire is not automatic but conditioned on the organizational infrastructure of the targeted movement, specifically the presence of campaign structure and parallel media institutions. Earl (2011) provides a complementary typology distinguishing iron fists from velvet gloves, arguing that the form of repressive action matters as much as its intensity.

These contributions highlights important variation in repression-mobilization dynamics. However, the gap remains: why similar, comparable repression events produce sustained collective action in some settings and permanent demobilization in others. This paper argues that spatial structure of the urban environment surrounding the protest site contribute to the diverging mechanism in within-type repression response variation.

2.2 Threshold dynamics and collective action

The threshold tradition in collective action provides the formalized approach in theorizing why repression has heterogeneous effects across individuals and populations. Granovetter (1978) develops the foundational model by assigning each individual a mobilization threshold representing the proportion of others who must already participate before they will join, and collective outcomes are determined by the distribution of thresholds across a population. Populations with nearly identical average preferences can produce different collective outcomes, because the shape of the threshold distribution and the seeding of the cascade condition the variation. Kuran (1989) develops a parallel model centered on preference falsification by modeling how individuals publicly support regimes they privately oppose, and observable mobilization can shift rapidly when a small shock reveals the previously hidden opposition.

Threshold models are nonetheless aspatial. They assume that every individual observes the same global proportion of mobilized others, and that this proportion drives the cascade. However, after a dispersal event, individuals observe not a global rate but a local one, conditioned on whoever happens to be in their immediate vicinity. Petersen (2001) shows in the context of Lithuanian resistance to Soviet occupation that the critical movement from passive opposition to organized participation is governed by threshold dynamics operating at the community level. It operates with community-based safety calculations and norms of reciprocity, rather than society-wide observation, determining whether individuals commit to organized collective action. The spatial argument developed here builds on this insight by treating the local observation environment as itself a variable, conditioned by built environment features.

2.3 Spatial structure in contentious politics

A growing body of work has argued that physical space is constitutive of contentious politics. Sewell (2001) argues that spatial configurations actively shape the repertoires of action available to contentious actors. The spatial logics of contention determine who can assemble where, how visible action is to third parties, and how state forces can or cannot control access.

Tilly (2000) makes a parallel argument that the spaces of contention are constitutive elements of political action, shaping which actors can coordinate and which repertoires become feasible. Scott (1998) argues that states systematically render environments legible to facilitate control, and that organic, irregular morphologies resist this legibility. This logic extends to the asymmetric mobility advantage protesters hold in dense kampung alleys relative to security forces operating in planned open space.

These spatial arguments have remained largely theoretical and ethnographic, with limited formalization. Wood (2003) and Viterna (2006) provide micro-foundational work that complements the spatial argument. Wood documents non-instrumental bases of participation that survive repressive episodes, establishing that dispersed individuals retain the motivation to remobilize. Viterna shows that situational co-presence with other committed actors is a necessary condition for remobilization to occur. The spatial mechanism formalized in this paper demonstrates that built environment determines whether this co-presence actually obtains after dispersal.

3. The Spatial Mechanism

3.1 Parameters and typologies

The mechanism this paper formalizes operates through two parameters that capture features of the urban built environment. The first parameter is **refuge density**, denoted ρ (rho) or density, which represents the probability that a dispersing individual can shelter in a given cell of the surrounding urban space. Dense kampung (urban village) morphology, with overlapping residential structures, narrow alleys, mosques, impermanent food stalls, and small commercial spaces, offers high refuge density. Dispersing individuals can quickly find cover, maintain proximity to the protest site, and encounter others doing the same. Planned commercial districts and open boulevards offer low refuge density, pushing dispersing individuals into exposed and legible space where continued presence is costly and encounter with co-mobilizers is unlikely. Refuge density therefore determines both physical safety after dispersal and the social density that sets the conditions for local threshold dynamics.

The second parameter is **permeability**, denoted p , which captures the probability that a dispersing individual can move through a given cell without interception. This parameter operationalizes the asymmetric mobility advantage that dense and irregular urban morphology confers on individual protesters relative to security forces. Narrow kampung alleys that can be navigated on foot but not by security vehicles produce high permeability for individuals and low effective permeability for state forces. Wide boulevards and open plazas invert this asymmetry since the open space makes them easily controlled by security forces and offer dispersing individuals few options beyond flight or surrender.

Refuge density and permeability are empirically correlated across the three typologies, but they enter the causal chain at different points and capture distinct features of built environment. In kampung morphology, both are high: dense residential structures provide cover while narrow alleys confer asymmetric mobility advantage. In planned open boulevard, both are low: exposed

sightlines eliminate shelter while wide roads give state forces full mobility. Mixed commercial morphology produces intermediate and variable outcomes because partial refuge density and partial permeability create pockets of favorable conditions without sustaining them across the grid. Agents who land in a shophouse alley find the same clustering advantage as kampung, while agents at an open intersection face the same exposure as open boulevard.

However, there is also potential of divergence. A private office district consists of dense building footprints produce high refuge density, but security personnel, locked gates, and surveillance systems reduce permeability to near-zero. However, a campus corridor is the opposite: it has many pedestrian pathways therefore has high permeability, but the building designed for circulation rather than concealment (low refuge density). The base model tests co-varying typologies to isolate the morphological mechanism broadly; decoupling refuge density and permeability in a secondary sweep would identify which feature does more causal work in enabling cascade.

3.2 Causal chain

The mechanism produces aggregate mobilization outcomes from individual-level threshold decisions through a clear causal chain. Each individual carries a personal mobilization threshold drawn from a population distribution. After repression triggers dispersal at L , each individual moves outward through cells whose permeability determines whether the escape attempt succeeds. Dispersal duration operationalizes repression intensity: longer duration means more sustained intervention, forcing individuals to keep moving before they can attempt shelter. The individual then attempts to shelter in their landing cell with success probability determined by refuge density. Once sheltered, the individual observes the local mobilization rate within a configurable observation area: the individual's own cell or the surrounding Moore neighborhood. Then the agent crosses into the mobilized state if this rate meets or exceeds their personal threshold. The aggregate mobilization outcome is the count of individuals who have crossed their thresholds at any given time. Built environment enters this chain at three points: escape, shelter, and observation. High refuge density and permeability together produce dense local observation environments where threshold crossings cascade through co-located peers in their own cell or in the surrounding neighborhood; low values produce isolated environments where individual threshold crossings exist as isolated events.

3.3 Interaction and prediction

The interaction of the two parameters determines whether dispersal is temporary or permanent. High refuge density and high permeability together, the kampung condition, enable individuals to escape, find shelter near one another, and encounter sufficient co-mobilizers to trigger local threshold cascades. Low refuge density and low permeability together, the open boulevard condition, prevent successful escape, expose individuals to re-interception, and scatter survivors too widely for cascade to ignite. The mixed commercial condition produces variable outcomes contingent on where agents land within the fragmented morphology. Two additional parameters condition cascade within any morphology: observation radius determines how wide an area sheltered agents scan for mobilized peers, and cell capacity operationalizes ongoing

state monitoring by re-dispersing agents when gatherings exceed a threshold size. The substantive prediction is that the rank order kampung greater than mixed greater than open holds across reasonable variation in threshold distributions and observation radius, demonstrating that spatial structure is generatively sufficient to produce systematic variation in mobilization outcomes.

4. Model Description

4.1 Agents and environment

The model represents a stylized urban environment as a 30 by 30 grid of cells with each cell carrying two morphological attributes: refuge density and permeability, both real-valued in the interval zero to one. The base experimental setup initializes the entire grid with a single environment typology, comparing three typologies across runs. Heterogeneous grids with mixed typologies are a natural extension held out of the base model to isolate the morphological mechanism.

Agents represent individual residents who may or may not participate in collective action. Each agent carries three attributes: a mobilization threshold drawn from a normal distribution with stipulated mean and standard deviation, a current state from a five-state machine, and a position on the grid. The five states are dormant (not currently engaged), mobilized (actively participating in collective action), dispersing (moving outward after repression), sheltered (found cover but not yet remobilized), and demobilized (gave up and went home or terminal state). Agents are initialized in the mobilized state and placed at protest site L, located near the center of the grid with small random jitter to avoid grid-geometry artifacts.

Repression is modeled as a spatial event. At step one, a flag is raised at location L that transitions all mobilized agents present there into the dispersing state. By holding state behavior fixed and exogenous, variation in outcomes across runs is attributable solely to differences in the surrounding built environment.

4.2 Model mechanism

The model unfolds in discrete time steps. At each step, four operations occur in sequence: repression is triggered if the current step equals the repression step; all agents are activated in randomized order to avoid activation-sequence bias; cell capacity is checked and enforced; and the data collector records state counts before the model checks stopping rules.

Dispersing agents execute three sub-operations each step. They select a target cell from the Moore neighborhood, preferring cells farther from L (outward bias) and cells with more co-dispersing or sheltered agents (social attraction, implemented as weighted random selection). The movement attempt succeeds with probability p evaluated at the target cell. After a fixed dispersal duration of three steps, the agent attempts to shelter with success permeability p evaluated at current position. The outcome would consist of success transitions to be sheltered, failure transitions to be demobilized.

Sheltered agents execute the threshold update each step under two rules in order. The instigator rule transitions agents whose threshold falls below the instigator threshold parameter directly to mobilized without requiring peer observation. The spatial threshold rule applies to all others: the agent computes the local mobilization rate across all agents in the observation area (own cell if observation_radius=0, Moore neighborhood if observation_radius=1) and transitions to mobilized if this rate meets or exceeds personal threshold.

Two extensions applied in this final iteration. Cell capacity (default 15) operationalizes ongoing state monitoring: when more than cell_capacity mobilized or sheltered agents occupy a single cell, the state re-disperses them, representing continued crowd suppression beyond the initial event at L. Observation radius is implemented as a parameter rather than a design choice, enabling direct comparison of cell-level and neighborhood-level observation in the batch sweep. The stopping condition applies when no agents remain in transitional states, or when the maximum step count is reached.

4.3 Iterative development

The model went through four substantive iterations, each addressing a specific computational problem. Parameters were held constant across versions to isolate the effect of mechanism changes.

Version 1 produced zero reconstitution across all typologies. All agents dispersed simultaneously. No mobilized peers for sheltered agents to observe, therefore no cascade seed existed. Version 2 introduced the instigator rule, solving the seeding problem in principle but producing only marginal reconstitution: independent random walks scattered instigators too thinly to seed a cascade. Version 3 introduced social attraction in dispersal movement, motivated by Wood (2003) and Viterna (2006), and adjusted threshold parameters to a regime where cascades can ignite given sufficient local seeding. Version 4 introduced cell capacity and observation radius as parameters. Cell capacity operationalizes ongoing state monitoring, observation radius enabling direct comparison of observation assumptions in the batch sweep. The primary claim across all versions is robustness of rank order across typologies, not the specific reconstitution magnitudes.

5. Analysis Design

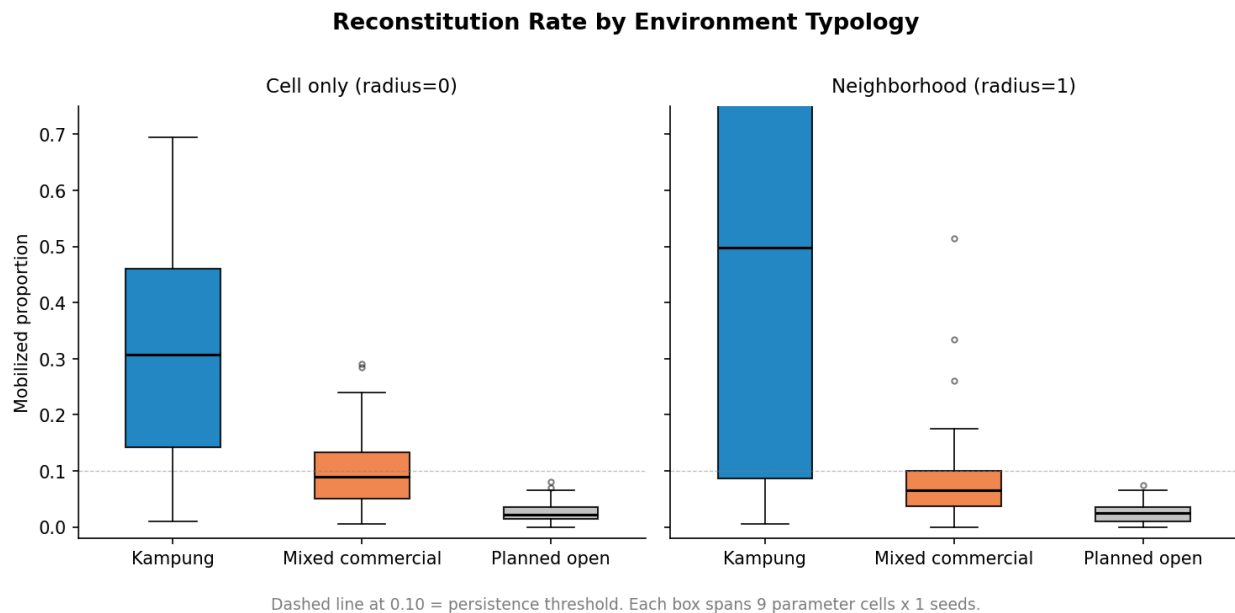
The batch sweep varies five dimensions: environment typology (kampung, mixed, open), threshold mean (0.25, 0.30, 0.35), threshold standard deviation (0.10, 0.15, 0.20), observation radius (0, 1), and random seed (ten replications per cell). This produces 540 runs total: 3 x 3 x 3 x 2 x 10. The primary outcome variable is the reconstitution rate: the proportion of agents in the mobilized state at the final time step. Secondary metrics include peak post-repression mobilization, time to peak, and persistence above ten percent of the population.

The primary hypothesis under test is that the rank order kampung greater than mixed greater than open holds across the threshold parameter space and across both observation radius. The threshold parameter sweep tests whether the rank order is specific to the default parameter

configuration or robust to reasonable variation in population threshold distributions. The observation radius comparison tests whether the result depends on whether agents observe their own cell or the surrounding neighborhood. The threshold parameter region most likely to produce violations is high threshold mean combined with low threshold standard deviation, because in this region cascade is rare enough in all typologies that stochastic variation can overwhelm morphological differences.

Dispersal duration operationalizes repression intensity: a short duration represents a brief intervention where agents quickly attempt to shelter, while a longer duration represents sustained pursuit that forces individuals to keep moving. The base model holds dispersal duration fixed at three steps to isolate morphological effects, but varying it in a secondary sweep would test whether rank order holds under different repression intensities.

6. Results



The batch run results confirm the primary hypothesis. **Figure 1** presents reconstitution rate distributions across all 540 runs by typology and observation radius. Kampung consistently produces higher reconstitution than mixed, which consistently exceeds planned open, across both observation modes. At observation radius=0, mean reconstitution rates are kampung 0.31, mixed 0.10, open 0.03, a kampung-to-open ratio of 11.9x. At observation radius=1, rates are kampung 0.45, mixed 0.08, open 0.02, a ratio of 19.2x. The kampung distribution is wide in both panels, reflecting bimodal dynamics where the cascade either fully ignites (high reconstitution) or fails to ignite (near-zero), depending on stochastic clustering during dispersal. Mixed and open distributions are narrow and low, indicating more predictable near-zero outcomes.

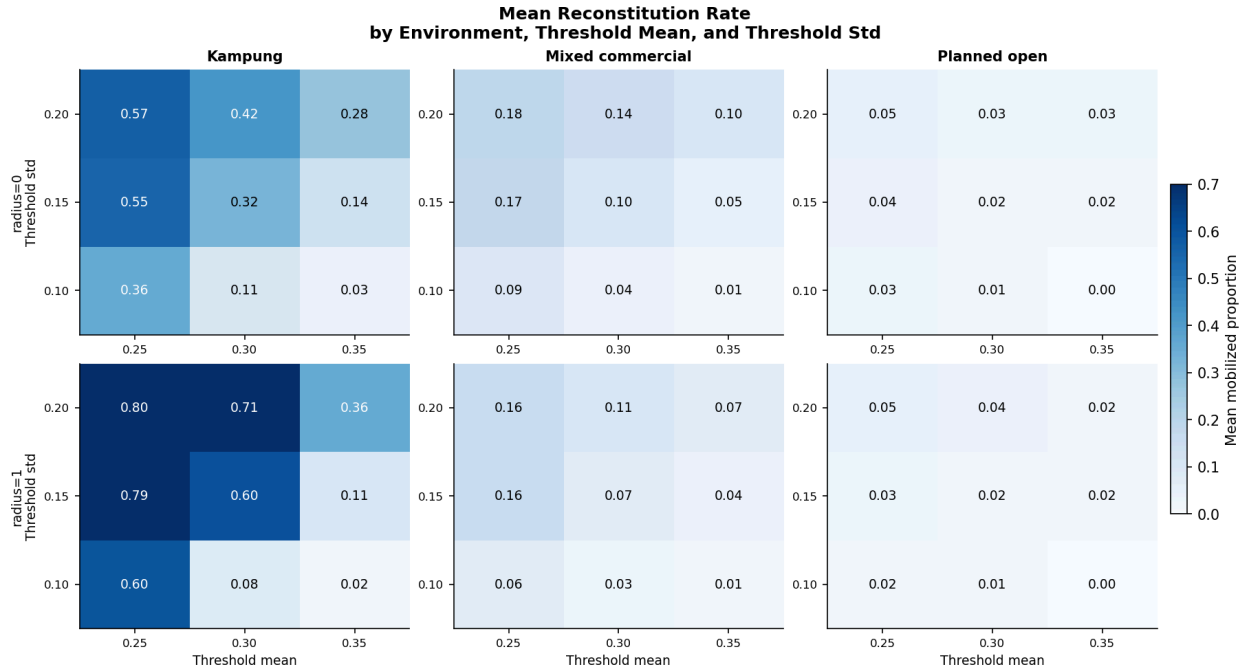


Figure 2 presents mean reconstitution rates per parameter cell for each typology, broken down by threshold parameters and observation radius. Three patterns are visible. First, threshold mean is the dominant parameter within each typology: kampung reconstitution falls from 0.57 at threshold mean=0.25 to 0.28 at threshold mean=0.35 (radius=0, threshold std=0.20), consistent with Granovetter's prediction that higher average thresholds suppress cascade. Second, observation radius amplifies kampung substantially, with mean rates at low threshold means rising from 0.57 to 0.80 (threshold std=0.20), 0.55 to 0.79 (threshold std=0.15), and 0.36 to 0.60 (threshold std=0.10). The same expansion produces a small reduction in mixed (mean 0.10 at radius=0 falling to 0.08 at radius=1) and leaves open effectively unchanged. This indicates that expanding the observation window to the surrounding neighborhood benefits typologies with sufficient post-dispersal density to provide observable mobilized peers; in sparse environments, the additional neighborhood cells contain mostly demobilized or absent agents, so the wider window supplies discouraging information. Third, the rank order between typologies is preserved on cell means across all parameter cells in both observation radius, though individual runs deviate in the boundary regime identified in Figure 3.

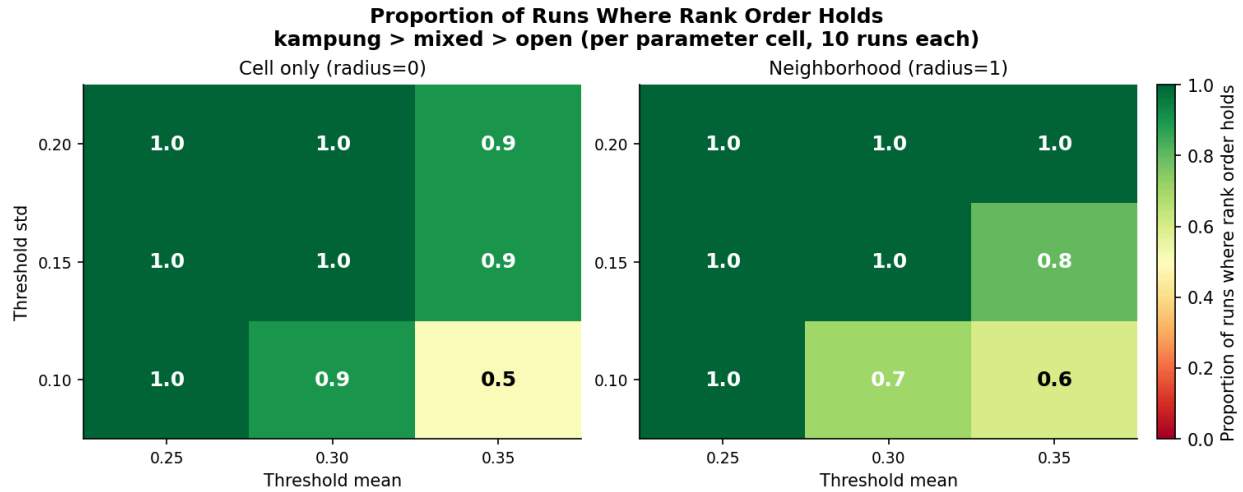


Figure 3 presents the proportion of runs (out of ten per parameter cell) in which the rank order kampung greater than mixed greater than open holds simultaneously. At observation radius=0, rank order holds in all ten runs for five of nine parameter cells, with the remaining cells ranging from 0.5 to 0.9 and the lowest value at threshold mean=0.35 and threshold standard deviation=0.10 (proportion=0.5). At observation radius=1, six of nine cells achieve a proportion of 1.0, with the three remaining cells at 0.6, 0.7, and 0.8, again concentrated at high threshold mean and low threshold standard deviation. These violations share a common location in the parameter space, the high-mean low-variance corner where mean reconstitution in all typologies is near-zero (at threshold mean=0.35 and threshold standard deviation=0.10, kampung=0.03, mixed=0.01, open=0.00 under radius=0). This corner is a boundary condition at which individual-level stochasticity exceeds morphological differences, because cascade is too rare in any typology to produce consistent sequences across runs.

One counterintuitive implication of the mechanism is reduced permeability does not straightforwardly reduce cascade potential. It shortens the distance agents travel before attempting shelter, which leaves them closer to the dispersal site L and closer to one another. In a morphology without cell capacity constraints, very low permeability would paradoxically increase local clustering near the dispersal site and could accelerate cascade. The cell capacity parameter counteracts this dynamic: when too many agents concentrate in a single cell, the state re-disperses them, suppressing cascade even in otherwise favorable morphologies. This demonstrates how the state exploiting crowd concentration, returning to break up visible gatherings before threshold dynamics can ignite, and it ties the permeability effect to the dispersal duration limit. When movement is harder, cascade dynamics become more sensitive to state monitoring intensity than to morphology alone.

Across the parameter sweep, the rank order holds in at least eight of ten runs in 15 of 18 parameter cells (83 percent) and in 163 of 180 paired runs overall (91 percent), exceeding the 80 percent threshold specified in the analysis design and supporting the generative sufficiency claim.

7. Conclusion and Scope Conditions

The batch sweep across 540 runs confirms the primary hypothesis: urban morphology, operationalized as refuge density and permeability, is generatively sufficient to produce systematic variation in mobilization outcomes after state repression disperses a crowd. The rank order kampung greater than mixed greater than open holds on parameter-cell means in all eighteen cells, in 163 of 180 paired runs (91 percent), and in at least eight of ten runs in fifteen of eighteen cells (83 percent), exceeding the pre-registered 80 percent criterion.

Two findings extend beyond the primary result. First, observation radius interacts with morphology rather than operating uniformly. It is widening the perceptual field, amplifies reconstitution in kampung while slightly suppressing it in mixed commercial environments. I suspect because in more sparse post-dispersal settings, a wider window returns mostly demobilizing information, which discouraging reconstitution. Second, reconstitution in kampung is bimodal and contingent, depending on whether stochastic clustering during dispersal produces the initial threshold crossings that seed cascade, which means morphology sets the probability of ignition rather than guaranteeing it. The model establishes that any complete account of repression's mobilizational consequences must treat built environment as a systematic variable.

Four scope conditions bound these claims. State behavior is partially endogenized through the cell capacity parameter, which operationalizes ongoing monitoring, but the model does not include an adaptive state agent that selects targets or adjusts repression intensity strategically. The model contains no organizational mediation, so mobilization is unmediated threshold crossing among individuals rather than coordinated collective action through social movement organizations or leadership networks. The three typologies are simplified representations rather than calibrations to any specific neighborhood, so results demonstrate sufficiency of the mechanism rather than its empirical magnitude in a given city. Finally, observation radius in this model is stipulated rather than measured, yet the finding that perceptual range conditions the direction of mobilization is theoretically substantial. Therefore, future extension must ground this parameter empirically, through both qualitative and/or quantitative inquiry on how dispersed protesters actually perceive the dispersal experience across different urban settings and repressive contexts.

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